

BI-DIRECTIONAL PIG SIGNALLER

TYPE: **MECHANICAL INTRUSIVE**

MODEL: **LS2-TQZ**

A bi-directional intrusive pig signaller is a pipeline device that detects the passage of a pipeline pig (a device used for cleaning or inspecting pipelines) regardless of its direction of travel. It features an internal trigger mechanism that interacts with the passing pig, confirming its presence at a specific point in the line. The "bi-directional" aspect ensures reliable detection even if the signaller is installed in the wrong orientation, providing a robust and dependable signal to operators.



BI-DIRECTIONAL DETECTION

Reliable detection in both directions.



RUGGED & RELIABLE

Built for high pressure, harsh and hazardous service.



HIGH PRESSURE RATING

Up to 25 bar
(375 psi).



MULTIPLE OUTPUT OPTIONS

Mechanical flag, electrical switch or wireless alert.



EASY RESET

Manual reset mechanism.



HAZARDOUS AREA CERTIFIED

Ex d IIC T6 for explosive atmospheres.

01 GENERAL DESCRIPTION

The LS2-TQZ Bi-Directional Intrusive Pig Signaller is a robust and reliable pipeline accessory designed to detect, signal, and confirm the passage of pipeline pigs in both directions. It provides visual, electrical, or wireless confirmation, ensuring smooth pigging operations in oil, gas, water, and chemical pipelines.

The intrusive trigger mechanism ensures positive activation, even at high pressure and variable flow conditions, while its weatherproof and explosion-proof construction makes it ideal for hazardous industrial environments.

02 KEY FEATURES

- ✓ **Bi-Directional Detection** – Compatible with forward and reverse pig runs.
- ✓ **High Pressure Rating** – Suitable for pipelines up to 25 bar (375 psi).
- ✓ **Rugged Construction** – Stainless steel wetted parts for maximum durability.
- ✓ **Multiple Output Options** – Mechanical flag indicator, electrical switch, or wireless alert.
- ✓ **Intrusive Trigger Design** – Ensures reliable mechanical activation.
- ✓ **Easy Reset** – Manual or automatic reset mechanisms available.
- ✓ **Hazardous Area Certification** – Ex d IIC T6 for explosive atmospheres.

05 ORDERING INFORMATION

MODEL	TYPE	OUTPUT	RESET	MATERIAL
LS2-TQZ	Intrusive Bi-directional	Mechanical Flag	Manual	Stainless Steel

Note: Other output types (Electrical, Wireless) and materials available on request.

03 TECHNICAL SPECIFICATIONS

PARAMETER	SPECIFICATION
Pipeline Size	4" to 48" (Custom sizes available)
Pressure Rating	Up to 25 bar (ANSI Class 300)
Temperature Range	-20°C to +120°C (-4°F to 248°F)
Body Material	Stainless Steel (SS316 on request)
Wet Parts	SS316
Seal Material	Viton / PTFE
Indicator Type	Mechanical flag
Process Connection	2" NPT / Flanged as per ASME B16.5
Certification	ATEX / IECEx Zone 1 & 2
Reset Option	Manual
Standoff Length	172 – 177 mm
Trigger Insertion	21 – 22 mm (Inside pipe)
Nozzle Height	As per Standoff length
Detection Capability	Designed to detect Sphere, steel body pigs, polyurethane pigs

04 TYPICAL APPLICATIONS



Oil & Gas
Transmission Pipelines



Product Batching
Lines



Chemical &
Petrochemical Lines



Water & Industrial
Utility Pipelines



06 STANDARD COMPLIANCE

- ASME B31.4 / B31.8
- API 6D and API RP 1130
- ATEX / IECEx (for hazardous installations)